

An Abstract
On
**AI-Based Personal Finance Manager with Smart Expense
Prediction System**

By

S. HEMANTH KUMAR

224G1A0528

B. JAHNAV ARJUN GOWD

224G1A0532

T. PAVAN

224G1A0567

Under the esteemed guidance of

Mr. P. Shajahan, M.Tech
Assistant Professor



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

**SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)
ANANTHAPURAMU**

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Project Coordinator

Head of the Department

AI-Based Personal Finance Manager with Smart Expense Prediction System

ABSTRACT

This project presents an AI-Based Personal Finance Manager with a Smart Expense Prediction System designed to improve financial planning and support better expense management. The system is implemented as a web-based application using the Django framework, enabling users to track daily expenses, manage category-wise budgets, monitor savings goals, and analyze overall spending patterns through an interactive dashboard.

The core of the system is a machine learning-based prediction engine that utilizes the XGBoost Regression model to forecast future expenses based on historical financial data and user spending behavior. The model learns complex patterns from past transactions and generates expense predictions when requested by the user through the application interface. These predictions help users understand possible future expenditures and identify potential budget risks.

In addition, the platform includes features such as automated budget monitoring, anomaly detection for unusual spending behavior, savings goal tracking, and personalized financial insights. The integration of predictive analytics with real-time expense tracking improves user awareness and supports informed financial decision-making. Overall, the proposed system transforms traditional expense tracking into an intelligent, data-driven financial management solution that promotes disciplined budgeting and long-term financial stability.

PROJECT GUIDE

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